DIGITAL SAT MATH PRACTICE

Quadratic Word Problems — SOLUTIONS

COMPLETE SOLUTIONS WITH TRAP ALERTS AND STRATEGY

Question 1 [Easy]

Correct Answer: A

Step by Step:

Step 1: The vertex gives the maximum. For $h(t) = -16t^2 + 48t$, use $t = -b/(2a)$.

Step 2: $t = -48/(2(-16)) = -48/(-32) = 1.5$ seconds.

Step 3: $h(1.5) = -16(1.5)^2 + 48(1.5) = -16(2.25) + 72 = -36 + 72 = 36$ feet.

Why Each Wrong Option Fails:

Option B: 48 is the coefficient of t (initial velocity), not the height.

Option C: 64 comes from a miscalculation like $-16 + 48 = 32$, doubled.

Option D: $96 = 48 \times 2$. Confuses doubling the velocity with finding height.

■ TRAP ALERT: 48 looks like it could be the height since it's the bigger coefficient, but it's the initial velocity.

■ KEY INSIGHT: For $h = at^2 + bt + c$, the vertex time is $t = -b/(2a)$. Always plug back in to find the max height.

■ FAST METHOD: $t = 48/32 = 1.5$. Then $h = -36 + 72 = 36$.

■✍■ VERDICT: Desmos is faster. Type the equation and read the vertex from the graph.

Question 2 [Easy]

Correct Answer: A

Step by Step:

Step 1: Width = w, Length = w + 5.

Step 2: Area = length × width = $w(w + 5)$.

Step 3: Set equal to 84: $w(w + 5) = 84$.

Why Each Wrong Option Fails:

Option B: Uses $w - 5$, but the length is 5 MORE, not 5 less.

Option C: This is a perimeter equation, not area.

Option D: This just adds the two sides without multiplying.

■ TRAP ALERT: Choice C sets up a perimeter equation. Area = length × width, NOT 2l + 2w.

■ KEY INSIGHT: "More than" means addition. Length = w + 5. Area = product of dimensions.

■ FAST METHOD: Width = w, length = w + 5, area = product = $w(w+5) = 84$.

■✍■ VERDICT: Hand solve is faster. This is equation setup only, no solving needed.

Question 3 [Easy]

Correct Answer: C

Step by Step:

Step 1: The rocket returns to ground when $h(t) = 0$.

Step 2: $-5t^2 + 30t = 0$.

Step 3: Factor: $-5t(t - 6) = 0$.

Step 4: $t = 0$ or $t = 6$. Since $t = 0$ is launch, the answer is $t = 6$.

Why Each Wrong Option Fails:

Option A: 3 is the time at the vertex (maximum height), not when it lands.

Option B: 5 is the absolute value of the leading coefficient.

Option D: 10 doesn't correspond to any feature of this equation.

■ TRAP ALERT: $t = 3$ is the time of max height (vertex), not the landing time. Read what the question asks!

■ KEY INSIGHT: Landing = $h(t) = 0$. Factor out t: t(coefficient stuff) = 0. The nonzero root is the landing time.

■ FAST METHOD: Set $h = 0$: $-5t(t-6) = 0$. Nonzero root = 6.

■✍■ VERDICT: Hand solve is faster. Quick factoring. Desmos confirms but is slower to set up.

Question 4 [Easy]

Correct Answer: B

Step by Step:

Step 1: Let the integers be n and $n + 1$.

Step 2: $n(n + 1) = 156$.

Step 3: $n^2 + n - 156 = 0$.

Step 4: Factor: $(n - 12)(n + 13) = 0$.

Step 5: $n = 12$ or $n = -13$. Since n is positive, $n = 12$.

Why Each Wrong Option Fails:

Option A: $11 \times 12 = 132 \neq 156$.

Option C: 13 is the larger of the two consecutive integers (12 and 13).

Option D: $14 \times 15 = 210 \neq 156$.

■ TRAP ALERT: Choice C gives the LARGER integer. The question asks for the SMALLER one.

■ KEY INSIGHT: Check: $12 \times 13 = 156$. ✓ Always verify your answer fits the problem.

■ FAST METHOD: Estimate: $\sqrt{156} \approx 12.5$, so try $12 \times 13 = 156$. ✓

■✍■ VERDICT: Hand solve is faster. Estimation gives the answer almost instantly.

Question 5 [Easy]

Correct Answer: C

Step by Step:

Step 1: $f(0) = -16(0)^2 + 64(0) + 3 = 3$.

Step 2: When $t = 0$ (at launch), the height is 3 feet.

Step 3: The constant term c in $h = at^2 + bt + c$ is always the initial height.

Why Each Wrong Option Fails:

Option A: Max height requires finding the vertex, not reading the constant.

Option B: Landing time requires solving $h(t) = 0$.

Option D: Speed is related to the coefficient of t (here, 64), not the constant.

■ **TRAP ALERT: Students sometimes confuse the constant term with the maximum height.**

■ **KEY INSIGHT: In any projectile equation $h = at^2 + bt + c$, c = initial height (height at $t = 0$).**

■ **FAST METHOD: Plug in $t = 0$ to identify the constant: $f(0) = 3$ = initial height.**

■✍■ **VERDICT: Hand solve is faster. Conceptual question, no computation needed.**

Question 6 [Medium]

Correct Answer: D

Step by Step:

Step 1: Let the width perpendicular to the barn be w. The fencing covers 2 widths + 1 length.

Step 2: $2w + L = 120$, so $L = 120 - 2w$.

Step 3: Area = $w \times L = w(120 - 2w) = 120w - 2w^2$.

Step 4: Vertex: $w = -120/(2(-2)) = 120/4 = 30$.

Step 5: $A(30) = 120(30) - 2(900) = 3600 - 1800 = 1800$.

Why Each Wrong Option Fails:

Option A: $900 = 30^2$. This squares the width but doesn't compute the actual area.

Option B: 1200 would result from using $w = 20$ instead of 30.

Option C: 1500 would result from an incorrect perimeter setup with all 4 sides.

■ **TRAP ALERT: The barn replaces one side, so perimeter = $2w + L$ (not $2w + 2L$). Using $2w + 2L = 120$ gives the wrong answer (900).**

■ **KEY INSIGHT: One side is free (barn wall): use $2w + L$ = fencing. This changes the optimization.**

■ **FAST METHOD: $w = 120/4 = 30$, $L = 60$. Area = $30 \times 60 = 1800$.**

■✍■ **VERDICT: Hand solve is faster. Standard optimization with clear formula.**

Question 7 [Medium]

Correct Answer: B

Step by Step:

Step 1: The vertex x-coordinate gives the number of units for max profit.

Step 2: $x = -b/(2a) = -100/(2(-2)) = -100/(-4) = 25$.

Why Each Wrong Option Fails:

Option A: 20 is a distractor that doesn't come from the vertex formula.

Option C: 30 might come from computing 300/10 or another arithmetic error.

Option D: $50 = 100/2$. This divides b by 2 but forgets the a coefficient.

■ **TRAP ALERT: Choice D divides b by 2 only, ignoring the denominator's a. The formula is $-b/(2a)$, not $b/2$.**

■ **KEY INSIGHT: Vertex formula: $x = -b/(2a)$. Remember the negative sign and both coefficients.**

■ **FAST METHOD: $-100/(2 \times -2) = -100/-4 = 25$.**

■✍■ **VERDICT: Hand solve is faster. One formula application.**

Question 8 [Medium]

Correct Answer: C

Step by Step:

Step 1: Set $h(t) = 0$: $-16t^2 + 32t + 48 = 0$.

Step 2: Divide by -16: $t^2 - 2t - 3 = 0$.

Step 3: Factor: $(t - 3)(t + 1) = 0$.

Step 4: $t = 3$ or $t = -1$. Since time is positive, $t = 3$ seconds.

Why Each Wrong Option Fails:

Option A: $h(1) = -16 + 32 + 48 = 64 \neq 0$.

Option B: $h(2) = -64 + 64 + 48 = 48 \neq 0$.

Option D: $h(4) = -256 + 128 + 48 = -80$ (already underground).

■ TRAP ALERT: The negative root ($t = -1$) is extraneous. Always take the positive time.

■ KEY INSIGHT: Divide by the GCF first to simplify factoring. Here, divide by -16.

■ FAST METHOD: $t^2 - 2t - 3 = (t-3)(t+1)$. Positive root = 3.

■✍■ VERDICT: Hand solve is faster. Clean factoring after simplifying.

Question 9 [Medium]

Correct Answer: A

Step by Step:

Step 1: Vertex: $x = -36/(2(-3)) = -36/(-6) = 6$.

Step 2: $R(6) = -3(36) + 36(6) - 5 = -108 + 216 - 5 = 103$.

Step 3: Since R is in thousands, max revenue = \$103,000.

Why Each Wrong Option Fails:

Option B: $108 = 216/2$. Forgets to subtract the other terms.

Option C: 103.5 results from a computation error in the arithmetic.

Option D: 105 is a rounded distractor.

■ TRAP ALERT: Don't forget the constant term (-5) when computing the vertex value. $-108 + 216 = 108$, then $108 - 5 = 103$.

■ KEY INSIGHT: Always plug the vertex x-value back into the FULL equation, including the constant.

■ FAST METHOD: $x = 6$, then $R = -108 + 216 - 5 = 103$. In thousands: \$103,000.

■✍■ VERDICT: Desmos is faster. Type the function and read the vertex y-value.

Question 10 [Medium]

Correct Answer: A

Step by Step:

Step 1: Perimeter $= 2w + 2L = 60$, so $w + L = 30$, giving $L = 30 - w$.

Step 2: Area $= w \times L = w(30 - w)$.

Why Each Wrong Option Fails:

Option B: Uses $L = 60 - w$, which forgets to divide the perimeter by 2.

Option C: Uses $L = 60 - 2w$, a barn wall setup (3 sides), not a full rectangle.

Option D: Uses half the semi-perimeter for some reason.

■ TRAP ALERT: The most common error is forgetting that $P = 2(w+L)$, so $w + L = P/2$, not P.
■ KEY INSIGHT: Perimeter = $2(w + L) \to w + L = 30 \to L = 30 - w$. Then Area = wL.
■ FAST METHOD: Halve the perimeter: $60/2 = 30$. $L = 30 - w$. Area = $w(30 - w)$.
■✍■ VERDICT: Hand solve is faster. Equation setup only.

Question 11 [Medium]

Correct Answer: 1.8

Step by Step:

Step 1: Set $h(t) = 0$: $-4.9t^2 + 3.5t + 10 = 0$.

Step 2: Multiply by -1: $4.9t^2 - 3.5t - 10 = 0$.

Step 3: Quadratic formula: $t = (3.5 \pm \sqrt{(12.25 + 196)})/9.8$.

Step 4: $= (3.5 \pm \sqrt{208.25})/9.8 = (3.5 \pm 14.43)/9.8$.

Step 5: Positive root: $t = (3.5 + 14.43)/9.8 = 17.93/9.8 \approx 1.83$.

Step 6: Rounded to the nearest tenth: 1.8.

■ TRAP ALERT: Don't forget to take only the positive root. Negative time is physically meaningless.
■ KEY INSIGHT: When dealing with gravity in meters, $a = -4.9$ (half of 9.8 m/s^2). In feet, $a = -16$.
■ FAST METHOD: Go straight to the quadratic formula or graph in Desmos.
■✍■ VERDICT: Desmos is faster. The decimals are messy. Graph and find the positive x-intercept.

Question 12 [Medium]

Correct Answer: A

Step by Step:

Step 1: Set $P(x) = 0$: $-5x^2 + 200x - 1500 = 0$.

Step 2: Divide by -5: $x^2 - 40x + 300 = 0$.

Step 3: Factor: $(x - 10)(x - 30) = 0$.

Step 4: $x = 10$ or $x = 30$.

Step 5: The FIRST break even point is $x = 10$.

Why Each Wrong Option Fails:

Option B: 15 is between the two roots, not a root itself.

Option C: 20 is the vertex (max profit price), not a break-even price.

Option D: 30 is the second break-even point, not the first.

■ TRAP ALERT: Both $x = 10$ and $x = 30$ are break-even points, but the question asks for the FIRST one.
■ KEY INSIGHT: Break even = $P(x) = 0$ = finding the roots. "First time" means the smaller root.
■ FAST METHOD: Divide by -5, then factor $x^2 - 40x + 300 = (x-10)(x-30)$. Smaller root = 10.
■✍■ VERDICT: Hand solve is faster. Clean factoring after dividing out the common factor.

Question 13 [Medium]

Correct Answer: C

Step by Step:

Step 1: The building height is the height at $t = 0$ (before the ball is thrown).

Step 2: $h(0) = -16(0)^2 + 32(0) + 80 = 80$ feet.

Why Each Wrong Option Fails:

Option A: 32 is the initial velocity coefficient, not the height.

Option B: 48 might come from some arithmetic with 16 and 32.

Option D: 96 = 80 + 16. Adds the leading coefficient to the constant.

■ **TRAP ALERT: Students sometimes add the vertex height to the initial height. The building height is simply $h(0) = c$.**

■ **KEY INSIGHT: The constant term in a projectile equation is always the initial height (height at $t = 0$).**

■ **FAST METHOD: Read the constant term: 80. That's the building height.**

■✍■ **VERDICT: Hand solve is faster. No computation needed. Read the constant.**

Question 14 [Medium]

Correct Answer: 160

Step by Step:

Step 1: Plug in $t = 5$: $N(5) = 2(25) + 12(5) + 50$.

Step 2: $= 50 + 60 + 50 = 160$.

■ **TRAP ALERT: Be careful with order of operations: $2t^2$ means $2(t^2)$, not $(2t)^2$.**

■ **KEY INSIGHT: Direct substitution. Evaluate each term separately to avoid errors.**

■ **FAST METHOD: 50 + 60 + 50 = 160. Enter 160.**

■✍■ **VERDICT: Hand solve is faster. Simple arithmetic substitution.**

Question 15 [Medium]

Correct Answer: B

Step by Step:

Step 1: Set $V(t) = 0$: $2t^2 - 80t + 800 = 0$.

Step 2: Divide by 2: $t^2 - 40t + 400 = 0$.

Step 3: Factor: $(t - 20)^2 = 0$.

Step 4: $t = 20$.

Why Each Wrong Option Fails:

Option A: 10 is half the answer.

Option C: 30 doesn't correspond to any feature.

Option D: 40 = 80/2. Divides the b coefficient by 2, which is the vertex formula, not the root.

■ **TRAP ALERT: This is a perfect square trinomial. $(t-20)^2 = 0$ has only one solution (double root).**

■ **KEY INSIGHT: A double root means the pool hits exactly zero and starts refilling (or the model ends).**

■ **FAST METHOD: $t^2 - 40t + 400 = (t-20)^2$. Answer: 20.**

■✍■ **VERDICT: Hand solve is faster. Recognize the perfect square pattern.**

Question 16 [Medium]

Correct Answer: C

Step by Step:

Step 1: The water lands when $h(x) = 0$.

Step 2: $-0.5x^2 + 4x = 0$.

Step 3: Factor: $x(-0.5x + 4) = 0$.

Step 4: $x = 0$ or $-0.5x + 4 = 0 \rightarrow x = 8$.

Step 5: $x = 0$ is the fountain. The water lands at $x = 8$.

Why Each Wrong Option Fails:

Option A: 4 is the vertex x-coordinate (maximum height location), not the landing point.

Option B: 6 doesn't correspond to any feature.

Option D: 10 comes from dividing 4 by 0.4 or some other error.

■ TRAP ALERT: $x = 4$ is where the water reaches max height (vertex), not where it lands. Zeros ≠ vertex.

■ KEY INSIGHT: The landing point is the nonzero root. Factor out x, then solve the remaining factor.

■ FAST METHOD: $4/0.5 = 8$. The nonzero root is 8.

■✍■ VERDICT: Hand solve is faster. Factor and divide.

Question 17 [Hard]

Correct Answer: B

Step by Step:

Step 1: Let x = number of \$1 increases.

Step 2: Price = $12 + x$. Tickets sold = $200 - 5x$.

Step 3: Revenue = $(12 + x)(200 - 5x)$.

Step 4: Expand: $R = 2400 - 60x + 200x - 5x^2 = -5x^2 + 140x + 2400$.

Step 5: Vertex: $x = -140/(2(-5)) = 140/10 = 14$.

Step 6: Price = $12 + 14$ = \$26.

Why Each Wrong Option Fails:

Option A: $22 = 12 + 10$. Uses $x = 10$ from some other calculation.

Option C: 30 uses $x = 18$ or confuses the ticket count with the price.

Option D: $32 = 12 + 20$. Uses $x = 20$ (the vertex of tickets sold, not revenue).

■ TRAP ALERT: The vertex x gives the NUMBER OF INCREASES, not the price. Add x back to the base price.

■ KEY INSIGHT: Revenue = Price × Quantity. Set up both as functions of x (the change variable).

■ FAST METHOD: Vertex of R: $x = 14$ increases. Price = $12 + 14$ = \$26.

■✍■ VERDICT: Desmos is faster for checking. Hand solve works well with the vertex formula.

Question 18 [Hard]

Correct Answer: 18

Step by Step:

Step 1: Let the smaller number be n. Then the larger is $n + 7$.

Step 2: $n(n + 7) = 198$.

Step 3: $n^2 + 7n - 198 = 0$.

Step 4: Factor: $(n - 11)(n + 18) = 0$.

Step 5: $n = 11$ (since $n > 0$).

Step 6: Larger number = 11 + 7 = 18.

■ TRAP ALERT: The question asks for the LARGER number. If you solve for n, you get 11, but you must add 7.

■ KEY INSIGHT: "Differ by 7" means the gap is 7. If smaller = n, larger = n + 7.

■ FAST METHOD: Estimate: √198 ≈ 14. Try 11 × 18 = 198. ✓ Larger = 18.

■✍■ VERDICT: Hand solve is faster. Estimation gets you there quickly. Enter 18.

Question 19 [Hard]

Correct Answer: A

Step by Step:

Step 1: Let W = width (shared direction), L = length.

Step 2: Fencing: 3W + 2L = 180, so L = (180 – 3W)/2 = 90 – 1.5W.

Step 3: Total area = W × L = W(90 – 1.5W) = 90W – 1.5W².

Step 4: Vertex: W = –90/(2(–1.5)) = 90/3 = 30.

Step 5: A(30) = 90(30) – 1.5(900) = 2700 – 1350 = 1350 sq ft.

Why Each Wrong Option Fails:

Option B: 1800 = 90 × 20. Uses incorrect vertex width.

Option C: 2025 = 45². Assumes a single pen with 2 sides.

Option D: 2700 = 90 × 30. Computes only the 90W term, forgetting the –1.5W² term.

■ TRAP ALERT: The shared dividing wall means 3 parallel sides, not 2. Set up the perimeter as 3W + 2L.

■ KEY INSIGHT: Shared wall pens: 3W + 2L = total fencing. One extra parallel fence segment.

■ FAST METHOD: W = 30, L = 45. Total area = 30 × 45 = 1350.

■✍■ VERDICT: Hand solve is faster. Standard optimization problem.

Question 20 [Hard]

Correct Answer: B

Step by Step:

Step 1: Vertex: x = –2/(2(–0.01)) = –2/(–0.02) = 100.

Step 2: h(100) = –0.01(10000) + 2(100) + 1 = –100 + 200 + 1 = 101.

Why Each Wrong Option Fails:

Option A: 100 = –0.01(10000) + 200 = 100. Forgets the +1 initial height.

Option C: 200 = 2(100). Uses only the bt term.

Option D: 201 = 200 + 1. Drops the –0.01x² term entirely.

■ TRAP ALERT: Choice A forgets the +1 (initial height). Every term matters when computing the vertex value.

■ KEY INSIGHT: Plug the vertex x into the FULL equation, including the constant term.

■ FAST METHOD: x = 100, then –100 + 200 + 1 = 101.

■✍■ VERDICT: Desmos is faster. Graph and read the vertex.

Question 21 [Hard]

Correct Answer: B

Step by Step:

Step 1: Since $a = 0.5 > 0$, the parabola opens upward: vertex is a minimum.

Step 2: Vertex: $x = -(-20)/(2(0.5)) = 20/1 = 20$.

Step 3: $C(20) = 0.5(400) - 20(20) + 350 = 200 - 400 + 350 = 150$.

Why Each Wrong Option Fails:

Option A: 100 comes from computing 350 – 250 or some other arithmetic error.

Option C: 200 = 0.5(400). Uses only the first term at the vertex.

Option D: 250 = 350 – 100. Partial computation.

■ TRAP ALERT: a > 0 means minimum (upward parabola), not maximum. Know which way the parabola opens.

■ KEY INSIGHT: Positive leading coefficient = opens up = minimum. Negative = opens down = maximum.

■ FAST METHOD: x = 20. C = 200 – 400 + 350 = 150.

■✍■ VERDICT: Hand solve is faster. Clean arithmetic at the vertex.

Question 22 [Hard]

Correct Answer: B

Step by Step:

Step 1: Area of triangle = (1/2)(base)(height) = $(1/2)(x)(x + 3) = 44$.

Step 2: $x(x + 3) = 88$.

Step 3: $x^2 + 3x - 88 = 0$.

Step 4: Factor: $(x - 8)(x + 11) = 0$.

Step 5: $x = 8$ (positive).

Step 6: Longer leg = $x + 3 = 11$.

Why Each Wrong Option Fails:

Option A: 8 is the shorter leg, not the longer one.

Option C: 14 = 8 + 6, using 2 × 3 = 6 instead of 3.

Option D: 16 = 2 × 8. Doubles the shorter leg instead of adding 3.

■ TRAP ALERT: The question asks for the LONGER leg. x = 8 is tempting but it's the shorter leg.

■ KEY INSIGHT: Triangle area = (1/2)bh, not bh. Don't forget the 1/2!

■ FAST METHOD: x(x+3) = 88. Try x = 8: 8 × 11 = 88. ✓ Longer leg = 11.

■✍■ VERDICT: Hand solve is faster. Quick mental estimation.

Question 23 [Hard]

Correct Answer: B

Step by Step:

Step 1: Set $h(t) = 149$: $-16t^2 + 96t + 5 = 149$.

Step 2: $-16t^2 + 96t - 144 = 0$.

Step 3: Divide by –16: $t^2 - 6t + 9 = 0$.

Step 4: Factor: $(t - 3)^2 = 0$.

Step 5: $t = 3$ seconds.

Why Each Wrong Option Fails:

Option A: 2 seconds: $h(2) = -64 + 192 + 5 = 133 \neq 149$.

Option C: 4 seconds: $h(4) = -256 + 384 + 5 = 133 \neq 149$.

Option D: 5 seconds: $h(5) = -400 + 480 + 5 = 85 \neq 149$.

■ **TRAP ALERT: 149 is the maximum height. A double root means the rocket reaches this height exactly once (at the vertex).**

■ **KEY INSIGHT: If the quadratic = 0 gives a perfect square, the target height IS the vertex height.**

■ **FAST METHOD: Divide by −16 to get $t^2 - 6t + 9 = (t-3)^2$. Answer: t = 3.**

■✍■ **VERDICT: Hand solve is faster. Recognize perfect square after simplifying.**

Question 24 [Hard]

Correct Answer: 4

Step by Step:

Step 1: Since $a = 3 > 0$, vertex is a minimum.

Step 2: $t = -(-24)/(2(3)) = 24/6 = 4$.

■ **TRAP ALERT: Positive leading coefficient means minimum, not maximum.**

■ **KEY INSIGHT: Vertex formula works for minimum ($a > 0$) and maximum ($a < 0$) alike.**

■ **FAST METHOD: 24/6 = 4. Enter 4.**

■✍■ **VERDICT: Hand solve is faster. One step formula.**

Question 25 [Hard]

Correct Answer: B

Step by Step:

Step 1: Vertex: $x = -(-0.8)/(2(0.02)) = 0.8/0.04 = 20$.

Step 2: $h(20) = 0.02(400) - 0.8(20) + 15 = 8 - 16 + 15 = 7$.

Why Each Wrong Option Fails:

Option A: 5 comes from 15 – 10 or another arithmetic path.

Option C: 8 = 0.02(400). Uses only the first term.

Option D: 10 = 0.8 × 12.5 or some other partial computation.

■ **TRAP ALERT: With decimals, it's easy to make arithmetic errors. Work carefully or use Desmos.**

■ **KEY INSIGHT: The midpoint between the poles ($x = 20$) is where the cable sags lowest.**

■ **FAST METHOD: $x = 20$: 8 − 16 + 15 = 7.**

■✍■ **VERDICT: Desmos is faster with decimal coefficients. Graph to verify.**

Question 26 [Hard]

Correct Answer: B

Step by Step:

Step 1: New dimensions: $(10 + x)$ and $(10 - x)$.

Step 2: Area = $(10 + x)(10 - x) = 100 - x^2$.

Step 3: $100 - x^2 = 91$.

Step 4: $x^2 = 9$.

Step 5: $x = 3$ (positive value).

Why Each Wrong Option Fails:

Option A: If $x = 1$: $(11)(9) = 99 \neq 91$.

Option C: If $x = 5$: $(15)(5) = 75 \neq 91$.

Option D: If $x = 9$: $(19)(1) = 19 \neq 91$.

■ **TRAP ALERT: Recognize the difference of squares pattern: $(a+b)(a-b) = a^2 - b^2$.**

■ **KEY INSIGHT: This is a difference of squares in disguise. $100 - x^2 = 91 \rightarrow x^2 = 9$.**

■ **FAST METHOD: $100 - 91 = 9 = x^2 \rightarrow x = 3$.**

■✍■ **VERDICT: Hand solve is faster. Difference of squares makes this almost instant.**

Question 27 [Hard]

Correct Answer: B

Step by Step:

Step 1: Set $h(t) = 240$: $-16t^2 + 128t - 240 = 0$.

Step 2: Divide by -16: $t^2 - 8t + 15 = 0$.

Step 3: Factor: $(t - 3)(t - 5) = 0$. Roots: $t = 3$ and $t = 5$.

Step 4: The projectile is above 240 ft between $t = 3$ and $t = 5$.

Step 5: Duration $= 5 - 3 = 2$ seconds.

Why Each Wrong Option Fails:

Option A: 1 second would be the duration if the roots were 1 apart.

Option C: 3 is one of the roots, not the duration.

Option D: 4 is the vertex time, not the duration.

■ **TRAP ALERT: The duration is the DIFFERENCE between the two roots, not one of the roots themselves.**

■ **KEY INSIGHT: For "how long above height H": solve $h(t) = H$, then subtract the two roots.**

■ **FAST METHOD: $t^2 - 8t + 15 = (t-3)(t-5)$. Duration $= 5 - 3 = 2$.**

■✍■ **VERDICT: Hand solve is faster. Factor and subtract roots.**

Question 28 [Hard]

Correct Answer: C

Step by Step:

Step 1: The base is where $h(x) = 0$.

Step 2: $-0.004x^2 + 1.6x = 0$.

Step 3: $x(-0.004x + 1.6) = 0$.

Step 4: $x = 0$ or $x = 1.6/0.004 = 400$.

Step 5: Width $= 400 - 0 = 400$ feet.

Why Each Wrong Option Fails:

Option A: 200 is the vertex x-value (the midpoint of the arch), not the width.

Option B: 300 doesn't correspond to any feature.

Option D: 500 comes from an arithmetic error in the division.

■ **TRAP ALERT: The vertex x = 200 is the MIDPOINT, not the width. Width = distance between the two zeros.**

■ **KEY INSIGHT: Width of an arch = distance between the x-intercepts.**

■ **FAST METHOD: 1.6/0.004 = 400. Width = 400 feet.**

■✍■ **VERDICT: Hand solve is faster. Simple division once you set h = 0.**

Question 29 [Hard]

Correct Answer: C

Step by Step:

Step 1: The subscriber count declines after the vertex.

Step 2: Vertex: $t = -8/(2(-0.5)) = 8/1 = 8$.

Step 3: Maximum subscribers occur at $t = 8$. The count begins declining after $t = 8$.

Step 4: So the subscriber count first begins to decline during month 9.

Why Each Wrong Option Fails:

Option A: Month 6 is before the peak; subscribers are still growing.

Option B: Month 8 is when subscribers peak, not when they start declining.

Option D: Month 10 is already 2 months into the decline.

■ **TRAP ALERT: The vertex (t = 8) is where the MAX occurs. Decline begins in the NEXT month (month 9).**

■ **KEY INSIGHT: "Begin to decline" means the month AFTER the vertex, not the vertex month itself.**

■ **FAST METHOD: Vertex at t = 8. Decline starts at t = 9 (the next month).**

■✍■ **VERDICT: Hand solve is faster. Find vertex, add 1.**

Question 30 [Hard]

Correct Answer: C

Step by Step:

Step 1: Set $d(t) = 0$: $3t^2 - 18t + 24 = 0$.

Step 2: Divide by 3: $t^2 - 6t + 8 = 0$.

Step 3: Factor: $(t - 2)(t - 4) = 0$.

Step 4: $t = 2$ and $t = 4$.

Why Each Wrong Option Fails:

Option A: Only includes one root. Both roots are valid.

Option B: Only includes one root.

Option D: $t = 3$ is the vertex (minimum displacement), not a zero.

■ **TRAP ALERT: t = 3 is the vertex, not a root. Don't confuse the vertex with the zeros.**

■ **KEY INSIGHT: A quadratic with two distinct real roots has TWO times when displacement = 0.**

■ **FAST METHOD: $t^2 - 6t + 8 = (t-2)(t-4)$. Both roots valid.**

■✍■ **VERDICT: Hand solve is faster. Clean factoring.**

Question 31 [Hard]

Correct Answer: B

Step by Step:

Step 1: Vertex: $x = -(-3)/(2(0.005)) = 3/0.01 = 300$.

Step 2: $y(300) = 0.005(90000) - 3(300) + 500 = 450 - 900 + 500 = 50$.

Why Each Wrong Option Fails:

Option A: 25 comes from a decimal error.

Option C: $75 = 500 - 450 + 25$, a different arithmetic mistake.

Option D: $100 = 500 - 400$, partial calculation.

■ **TRAP ALERT: Decimal coefficients can cause arithmetic errors. Double check each term.**

■ **KEY INSIGHT: $a > 0$ means the parabola opens upward: the vertex is the LOWEST point.**

■ **FAST METHOD: $x = 300$. $y = 450 - 900 + 500 = 50$.**

■✍■ **VERDICT: Desmos is faster. Graphing handles decimal arithmetic cleanly.**

Question 32 [Hard]

Correct Answer: A

Step by Step:

Step 1: Set $P(x) = 0$: $-2x^2 + 120x - 1000 = 0$.

Step 2: Divide by -2: $x^2 - 60x + 500 = 0$.

Step 3: Factor: $(x - 10)(x - 50) = 0$.

Step 4: $x = 10$ and $x = 50$.

Step 5: Since the parabola opens downward ($a < 0$), $P > 0$ between the roots.

Step 6: Profitable for $10 < x < 50$.

Why Each Wrong Option Fails:

Option B: 5 and 55 are symmetric about $x = 30$ but don't satisfy the equation.

Option C: 20 and 40 are symmetric about 30 but $P(20) = -800 + 2400 - 1000 = 600 > 0$ (inside, not roots).

Option D: 15 and 45 don't satisfy $x^2 - 60x + 500 = 0$.

■ **TRAP ALERT: The company is profitable BETWEEN the roots (since $a < 0$). Outside the roots, profit is negative.**

■ **KEY INSIGHT: For downward parabolas, the function is positive BETWEEN its zeros.**

■ **FAST METHOD: $x^2 - 60x + 500 = (x-10)(x-50)$. Profitable: 10 to 50.**

■✍■ **VERDICT: Hand solve is faster. Factor the quadratic cleanly.**

Question 33 [Hard]

Correct Answer: 71

Step by Step:

Step 1: Set $h(x) = 0$: $-0.02x^2 + 1.4x + 2 = 0$.

Step 2: Multiply by -50: $x^2 - 70x - 100 = 0$.

Step 3: Quadratic formula: $x = (70 \pm \sqrt{4900 + 400})/2 = (70 \pm \sqrt{5300})/2$.

Step 4: $\sqrt{5300} = \sqrt{100 \times 53} = 10\sqrt{53} \approx 72.8$.

Step 5: $x = (70 + 72.8)/2 = 142.8/2 = 71.4 \approx 71$.

■ **TRAP ALERT: Remember to use the POSITIVE root only. The negative root gives a negative distance.**

■ **KEY INSIGHT: When coefficients are decimals, multiply through to clear them before using the formula.**

■ **FAST METHOD: Desmos: graph $y = -0.02x^2 + 1.4x + 2$ and find the positive x-intercept.**

■✍■ **VERDICT: Desmos is faster. The decimals and irrational roots make hand computation slow. Enter 71.**

Question 34 [Hard]

Correct Answer: B

Step by Step:

Step 1: Let width = w and length = L. LW = 180 and (L – 3)(W + 2) = 180.

Step 2: Test choice B: L = 18, W = 10. LW = 180. ✓

Step 3: (18 – 3)(10 + 2) = 15 × 12 = 180. ✓

Step 4: Both conditions satisfied.

Why Each Wrong Option Fails:

Option A: 12 × 15 = 180 but (12–3)(15+2) = 9 × 17 = 153 ≠ 180.

Option C: 9 × 20 = 180 but (20–3)(9+2) = 17 × 11 = 187 ≠ 180.

Option D: 6 × 30 = 180 but (30–3)(6+2) = 27 × 8 = 216 ≠ 180.

■ **TRAP ALERT: All four choices have area = 180, but only one ALSO satisfies the modified area condition.**

■ **KEY INSIGHT: On multiple choice, backsolving (testing answer choices) is often faster than algebra.**

■ **FAST METHOD: Test B: 18 × 10 = 180 and 15 × 12 = 180. Both work. Done.**

■✍■ **VERDICT: Hand solve is faster. Backsolving from answer choices is the fastest strategy.**

Question 35 [Hard]

Correct Answer: A

Step by Step:

Step 1: Set h(t) = 240: $-16t^2 + 128t = 240$.

Step 2: $-16t^2 + 128t - 240 = 0$.

Step 3: Divide by –16: $t^2 - 8t + 15 = 0$.

Step 4: Factor: (t – 3)(t – 5) = 0 → t = 3 and t = 5.

Step 5: The firework is above 240 ft between t = 3 and t = 5.

Why Each Wrong Option Fails:

Option B: t = 2 to t = 6 would correspond to h = 192 ft, not 240.

Option C: Uses one correct root (3) and one wrong root (6).

Option D: Uses one wrong root (2) and one correct root (5).

■ **TRAP ALERT: Students may compute –240/–16 = 15 correctly but factor wrong.**

■ **KEY INSIGHT: "Higher than" a value: solve h(t) = value, then the interval is between the two roots.**

■ **FAST METHOD: $t^2 - 8t + 15 = (t-3)(t-5)$. Interval: [3, 5].**

■✍■ **VERDICT: Hand solve is faster. Clean factoring gives exact roots.**

Question 36 [Hard]

Correct Answer: A

Step by Step:

Step 1: Since a = 0.1 > 0, vertex is a minimum.

Step 2: x = –(–4)/(2(0.1)) = 4/0.2 = 20.

Step 3: D(20) = 0.1(400) – 4(20) + 50 = 40 – 80 + 50 = 10.

Why Each Wrong Option Fails:

Option B: Gets x = 20 correct but miscalculates D(20) as 15.

Option C: Doubles the vertex x-value to get 40.

Option D: Halves the vertex x-value and miscalculates D(10) = 10 – 40 + 50 = 20.

■ **TRAP ALERT: With decimal coefficients, be careful with arithmetic. 0.1 × 400 = 40, not 4.**

■ **KEY INSIGHT: Vertex gives both the optimal speed AND the minimum defect count. Plug x back in.**

■ **FAST METHOD: x = 20. D = 40 – 80 + 50 = 10. Speed 20, min defects 10.**

■✍■ **VERDICT: Hand solve is faster. Clean arithmetic at the vertex.**

Question 37 [Hard]

Correct Answer: 27

Step by Step:

Step 1: The camera should fire at the vertex time.

Step 2: t = –40/(2(–16)) = 40/32 = 1.25 seconds.

Step 3: h(1.25) = –16(1.5625) + 40(1.25) + 2 = –25 + 50 + 2 = 27 feet.

Step 4: The question asks for the height: 27 feet.

■ **TRAP ALERT: The question asks for the height at max, not the time. Read what's being asked for the grid-in.**

■ **KEY INSIGHT: Vertex time: t = 1.25. Vertex height: h = 27. The answer is the height.**

■ **FAST METHOD: t = 40/32 = 1.25. h = –25 + 50 + 2 = 27. Enter 27.**

■✍■ **VERDICT: Desmos is faster. Graph and read the vertex coordinates.**

Question 38 [Hard]

Correct Answer: A

Step by Step:

Step 1: Find h(40): h(40) = $-(40)^2/200$ + 40 = –1600/200 + 40 = –8 + 40 = 32.

Step 2: Height at the tree = 32 yards. Tree height = 10 yards.

Step 3: Clearance = 32 – 10 = 22 yards.

Why Each Wrong Option Fails:

Option B: 12 = 32 – 20. Uses 20 instead of 10 for the tree height.

Option C: Miscalculates h(40) as 8 instead of 32.

Option D: 32 is the ball's height, not the clearance above the tree.

■ **TRAP ALERT: Choice D gives the ball's HEIGHT (32), not the clearance ABOVE the tree (32 – 10 = 22).**

■ **KEY INSIGHT: Clearance = ball height at tree position minus tree height. Two separate calculations.**

■ **FAST METHOD: h(40) = –8 + 40 = 32. Clearance = 32 – 10 = 22.**

■✍■ **VERDICT: Hand solve is faster. Direct substitution and subtraction.**

Question 39 [Hard]

Correct Answer: B

Step by Step:

Step 1: Vertex: $t = -6/(2(-0.3)) = 6/0.6 = 10$.

Step 2: 10 years after 2020 = 2030.

Why Each Wrong Option Fails:

Option A: 2028 is 8 years after 2020, before the peak.

Option C: 2032 is 12 years after, past the peak.

Option D: 2035 is 15 years after. Not related to any feature.

■ **TRAP ALERT: The answer is a YEAR, not the value of t. Add t to the base year (2020).**

■ **KEY INSIGHT: "t years after 2020" means the actual year = 2020 + t.**

■ **FAST METHOD: t = 6/0.6 = 10. Year = 2020 + 10 = 2030.**

■✍■ **VERDICT: Hand solve is faster. Vertex formula then add to base year.**

Question 40 [Hard]

Correct Answer: 196

Step by Step:

Step 1: Perimeter = $2L + 2W = 56$, so $L + W = 28$.

Step 2: $L = 28 - W$.

Step 3: $A = W(28 - W) = 28W - W^2$.

Step 4: Vertex: $W = -28/(2(-1)) = 14$.

Step 5: $A(14) = 28(14) - (14)^2 = 392 - 196 = 196$.

Step 6: The maximum area is 196 cm^2. (A 14×14 square.)

■ **TRAP ALERT: The maximum area rectangle with fixed perimeter is always a SQUARE. L = W = P/4.**

■ **KEY INSIGHT: For fixed perimeter, max area = $(P/4)^2 = (56/4)^2 = 14^2 = 196$.**

■ **FAST METHOD: Square: side = 56/4 = 14. Area = 14^2 = 196. Enter 196.**

■✍■ **VERDICT: Hand solve is faster. Shortcut: max area rectangle = square with side P/4.**

TOPIC SUMMARY: QUADRATIC WORD PROBLEMS

• Projectile Motion Template: $h(t) = at^2 + bt + c$. a = gravity (-16 ft/s^2 or -4.9 m/s^2), b = initial velocity, c = initial height.

• Vertex Formula: $t = -b/(2a)$. This gives either the maximum ($a < 0$) or minimum ($a > 0$). Always plug back in to find the vertex VALUE.

• Zeros (Roots): Set $h = 0$ and solve. For projectiles, the positive root is the landing time. Negative roots are physically meaningless.

• Constant Term: In $h = at^2 + bt + c$, the value c is always the initial value (height at $t = 0$).

• Revenue Optimization: Revenue = Price × Quantity. Set up both as functions of a change variable, then maximize.

• Area Optimization: Express area as a quadratic in one variable using the constraint (perimeter, fencing). The vertex gives the maximum area.

• Barn Wall / Shared Fence Problems: Count fence segments carefully. Barn wall → 3 sides. Divider → extra parallel segment.

• Break Even Points: Set Profit = 0. The roots tell you where profit switches from negative to positive (and back).

• Time Above a Height: Solve h(t) = target. Duration = larger root minus smaller root.

• Backsolving: On multiple choice, plugging answer choices into the problem can be faster than algebra.

• Extraneous Roots: When solving radical or squared equations, always check answers in the original equation.

• Desmos Strategy: Graph the quadratic and read vertex/zeros directly. Fastest for decimal coefficients.

• Grid In Tips: Read WHAT the question asks (height vs time, smaller vs larger number). The most common error is solving for the wrong quantity.